

A Minimal Geometric Model for Topological Mass Emergence

Anonymous

Abstract

We construct a compact field model in which inertial mass arises from the geometric coupling between a tensioned filament network and a dynamical time-flow field. The framework is formulated purely as a mathematical representation: no claim is made regarding ontological reality. A self-consistent Lagrangian density is shown to yield an emergent mass term that depends only on topological invariants of the underlying structure. The resulting scaling law reproduces the observed mass hierarchy pattern up to a multiplicative constant and yields a falsifiable prediction for curvature-dependent inertial shifts at sub-cosmological scales.

1 Introduction

Classical and quantum field theories treat mass as an intrinsic property of matter fields or as a parameter entering symmetry-breaking potentials. We explore here a minimal alternative: a purely geometric representation in which effective mass emerges from the interaction between a one-dimensional filament current $J^{\mu\nu}$ and a unit time-flow vector u^μ on a four-dimensional background manifold. This construction, while not intended as an ontological claim, offers a mathematically transparent way to visualize how inertia may appear as a measure of topological misalignment.

2 Minimal Lagrangian

The model uses four dynamical fields: a compact scalar phase $\theta(x)$, a time-flow vector u^μ satisfying $u^\mu u_\mu = -1$, an antisymmetric filament current $J^{\mu\nu}$ with $\nabla_\mu J^{\mu\nu} = 0$, and the induced metric $g_{\mu\nu}$. The Lagrangian density is

$$\begin{aligned} \mathcal{L} = & \frac{1}{2\kappa} R + \frac{f_\theta}{2} \nabla_\mu \theta \nabla^\mu \theta - \Lambda_\theta^4 (1 - \cos \theta) + \frac{c_1}{2} (\nabla_\mu u_\nu) (\nabla^\mu u^\nu) + \frac{c_2}{2} (\nabla_\mu u^\mu)^2 \\ & + \frac{\kappa_0}{4} J_{\mu\nu} J^{\mu\nu} + \frac{\kappa_2 \ell^2}{4} (\nabla_\alpha J_{\mu\nu}) (\nabla^\alpha J^{\mu\nu}) + \frac{g_h}{2} F_{\mu\nu}(\theta) J^{\mu\nu} + \lambda (u^\mu u_\mu + 1), \end{aligned} \quad (1)$$

where $F_{\mu\nu}(\theta) = \nabla_{[\mu} \nabla_{\nu]} \theta$ and λ enforces the normalization of u^μ . All coefficients are real constants chosen so that the quadratic terms are positive definite.

3 Emergent Inertial Term

Variation of $\mathcal{L}$ with respect to u^μ and θ produces a coupled system in which the divergence of u^μ acts as an effective potential for θ oscillations. Linearization around an aligned configuration $v^\mu \parallel u^\mu$ yields a small misalignment angle θ_4 defined by $\cos \theta_4 = v^\mu u_\mu$. The induced local inertial density takes the simple form

$$m_{\text{ind}}(x) \simeq T \sin^2 \theta_4(x), \quad (2)$$

where T is the filament tension scale fixed by c_1 . Hence inertia appears as the resistance of a tensioned network to deflection of its time-flow alignment.

4 Topological Mass Scaling

Topology enters through the conserved charge

$$Q = L_{\text{wind}} + L_{\text{link}} + W_{\text{writhe}}, \quad (3)$$

the sum of normalized winding, linking, and writhing numbers of the filament bundle. The effective mass for a localized excitation is then

$$m_{\text{eff}} = \frac{m_0}{Q} \exp(-\gamma_\chi \tau_\chi), \quad (4)$$

where $m_0 \sim T/\ell_f$ sets the base scale, τ_χ is a discrete torsion invariant, and γ_χ is a phenomenological damping constant. This form reproduces the observed hierarchy between light and heavy fermions when τ_χ varies by integer steps of order unity.

5 Predictive Handle

The model predicts a small deviation between inertial and gravitational mass in regions of high curvature. If $\mathcal{R}$ denotes the local Ricci scalar, then to leading order

$$\frac{m_{\text{inertial}}}{m_{\text{gravitational}}} \simeq 1 + \alpha_{\text{BW}} \frac{\mathcal{R}}{R_c}, \quad (5)$$

where α_{BW} is a dimensionless coupling derived from c_1/T and $R_c \sim 10^{-20} \text{ m}^{-2}$ is the curvature scale at which filament strain begins to influence the time-flow field. A measurable deviation of order 10^{-15} at this curvature would serve as a falsification test.

6 Discussion and Outlook

The SATO-Blockwave formalism, interpreted here as a mathematical model, provides a compact route from geometric assumptions to an explicit mass formula. Its strength lies in simplicity: the emergent mass term follows directly from the alignment geometry without

the need for symmetry breaking. The construction is compatible with general covariance and remains perturbatively stable under small curvature.

Further work should evaluate the stability of the u^μ sector beyond linear order and compute numerical spectra for specific topological classes. Whether or not nature employs this mechanism, the model demonstrates that topological coupling between a tensioned filament field and a time-flow foliation is sufficient to generate an inertial term of the correct form.

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References

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